

CLASSIFYING HANDWRITTEN DIGITS

➤ Obtaining and preparing the MNIST dataset

- The MNIST database (Modified National Institute of Standards and Technology database): The training dataset consists of handwritten digits from 250 different people
- In scikit-learn, the `fetch_openml` function downloads the MNIST dataset from OpenML (<https://www.openml.org/d/554>) as pandas DataFrame.

CLASSIFYING HANDWRITTEN DIGITS

➤ Equations

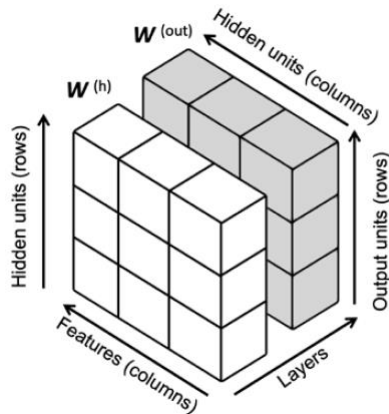
$$a^{(out)} = \begin{bmatrix} 0.1 \\ 0.9 \\ \vdots \\ 0.3 \end{bmatrix}, \quad y = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}$$

$$L(\mathbf{W}, \mathbf{b}) = \frac{1}{n} \sum_1^n \frac{1}{t} \sum_{j=1}^t (y_j^{[i]} - a_j^{(out)[i]})^2$$

$$\frac{\partial}{\partial \mathbf{w}_{j,l}^{(l)}} = L(\mathbf{W}, \mathbf{b})$$

$$\frac{d}{dx} [f(g(x))] = \frac{df}{dg} \cdot \frac{dg}{dx}$$

$$\frac{dF}{dx} = \frac{d}{dx} F(x) = \frac{d}{dx} f(g(h(u(v(x))))) = \frac{df}{dg} \cdot \frac{dg}{dh} \cdot \frac{dh}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}$$



$$\mathbf{Z}^{(h)} = \mathbf{X}^{(in)} \mathbf{W}^{(h)T} + \mathbf{b}^{(h)} \quad (\text{net input of the hidden layer})$$

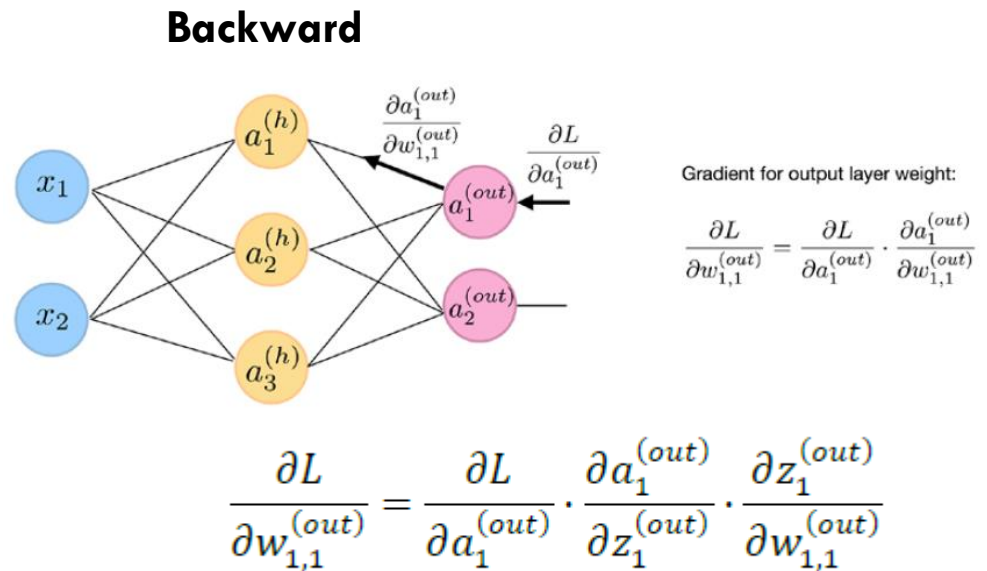
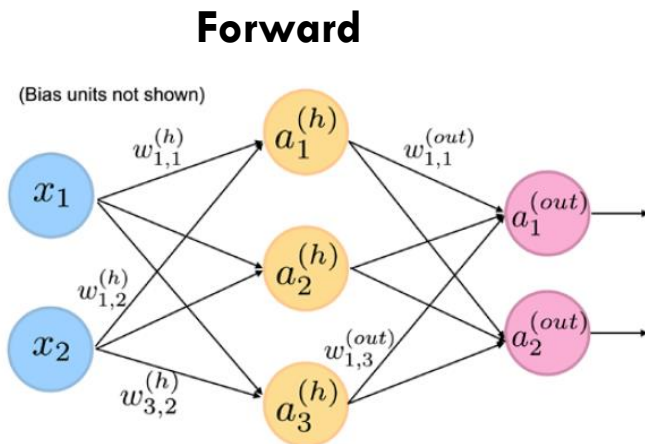
$$\mathbf{A}^{(h)} = \sigma(\mathbf{Z}^{(h)}) \quad (\text{activation of the hidden layer})$$

$$\mathbf{Z}^{(out)} = \mathbf{A}^{(h)} \mathbf{W}^{(out)T} + \mathbf{b}^{(out)} \quad (\text{net input of the output layer})$$

$$\mathbf{A}^{(out)} = \sigma(\mathbf{Z}^{(out)}) \quad (\text{activation of the output layer})$$

CLASSIFYING HANDWRITTEN DIGITS

➤ Forward and backward



CLASSIFYING HANDWRITTEN DIGITS

➤ Backpropagating the error of an NN

$$\frac{\partial L}{\partial a_1^{(out)}} = \frac{\partial}{\partial a_1^{(out)}} (y_1 - a_1^{(out)})^2 = 2(a_1^{(out)} - y)$$

$$\begin{aligned} \frac{\partial a_1^{(out)}}{\partial z_1^{(out)}} &= \frac{\partial}{\partial z_1^{(out)}} \frac{1}{1 + e^{z_1^{(out)}}} = \dots = \left(\frac{1}{1 + e^{z_1^{(out)}}} \right) \left(1 - \frac{1}{1 + e^{z_1^{(out)}}} \right) \\ &= a_1^{(out)} (1 - a_1^{(out)}) \end{aligned}$$

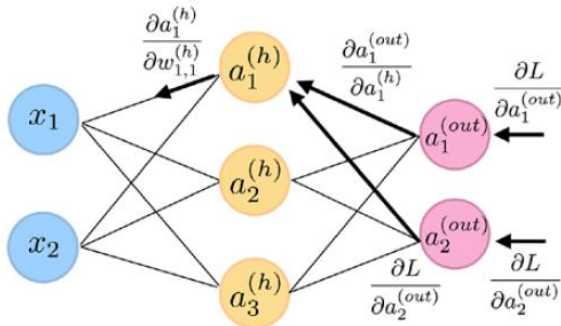
$$\frac{\partial z_1^{(out)}}{\partial w_{1,1}^{(out)}} = \frac{\partial}{\partial w_{1,1}^{(out)}} a_1^{(h)} w_{1,1}^{(out)} + b_1^{(out)} = a_1^{(h)}$$

$$\frac{\partial L}{\partial w_{1,1}^{(out)}} = \frac{\partial L}{\partial a_1^{(out)}} \cdot \frac{\partial a_1^{(out)}}{\partial z_1^{(out)}} \cdot \frac{\partial z_1^{(out)}}{\partial w_{1,1}^{(out)}} = 2(a_1^{(out)} - y) \cdot a_1^{(out)} (1 - a_1^{(out)}) \cdot a_1^{(h)}$$

$$w_{1,1}^{(out)} := w_{1,1}^{(out)} - \eta \frac{\partial L}{\partial w_{1,1}^{(out)}}$$

CLASSIFYING HANDWRITTEN DIGITS

➤ Backpropagating the error of an NN



Computing the partial derivatives of the loss with respect to the first hidden layer weight

Gradient for hidden layer weight:

$$\frac{\partial L}{\partial w_{1,1}^{(h)}} = \frac{\partial L}{\partial a_1^{(out)}} \cdot \frac{\partial a_1^{(out)}}{\partial a_1^{(h)}} \cdot \frac{\partial a_1^{(h)}}{\partial w_{1,1}^{(h)}} + \frac{\partial L}{\partial a_2^{(out)}} \cdot \frac{\partial a_2^{(out)}}{\partial a_1^{(h)}} \cdot \frac{\partial a_1^{(h)}}{\partial w_{1,1}^{(h)}}$$

$$\delta_1^{(out)} = \frac{\partial L}{\partial a_1^{(out)}} \cdot \frac{\partial a_1^{(out)}}{\partial z_1^{(out)}}$$

$$\frac{\partial L}{\partial w_{1,1}^{(h)}} = \frac{\partial L}{\partial a_1^{(out)}} \cdot \frac{\partial a_1^{(out)}}{\partial z_1^{(out)}} \cdot \frac{\partial z_1^{(out)}}{\partial a_1^{(h)}} \cdot \frac{\partial a_1^{(h)}}{\partial z_1^{(h)}} \cdot \frac{\partial z_1^{(h)}}{\partial w_{1,1}^{(h)}} + \frac{\partial L}{\partial a_2^{(out)}} \cdot \frac{\partial a_2^{(out)}}{\partial z_2^{(out)}} \cdot \frac{\partial z_2^{(out)}}{\partial a_1^{(h)}} \cdot \frac{\partial a_1^{(h)}}{\partial z_1^{(h)}} \cdot \frac{\partial z_1^{(h)}}{\partial w_{1,1}^{(h)}}$$

$$\frac{\partial L}{\partial w_{1,1}^{(h)}} = \delta_1^{(out)} \cdot \frac{\partial z_1^{(out)}}{\partial a_1^{(h)}} \cdot \frac{\partial a_1^{(h)}}{\partial z_1^{(h)}} \cdot \frac{\partial z_1^{(h)}}{\partial w_{1,1}^{(h)}} + \delta_2^{(out)} \cdot \frac{\partial z_2^{(out)}}{\partial a_1^{(h)}} \cdot \frac{\partial a_1^{(h)}}{\partial z_1^{(h)}} \cdot \frac{\partial z_1^{(h)}}{\partial w_{1,1}^{(h)}}$$